

AN EXPLICIT d^{-3} MOTION BOUND FOR PRIMITIVE DISTANCE-REGULAR GRAPHS

MACHINE-ASSISTED PROOF DEVELOPMENT

ABSTRACT. Let X be a primitive distance-regular graph on n vertices, of diameter $d \geq 3$. We prove, subject to the published inputs quoted precisely below, that either X is a Johnson graph or a Hamming graph, or [LLM]

$$\text{motion}(X) \geq \frac{500n}{5673d^3} = \frac{n}{11.346 d^3}. \quad [\text{LLM}]$$

The published Pyber–Skresanov argument gives a bound on the scale n/d^6 in the primitive case. [PS, Theorem 1.4] Four changes remove three powers of d and sharpen the constant: an exact formula for the vertices distinguishing an adjacent pair; use of the support-sensitive form of the geodesic boundary argument; retention of the full Metsch clique expression, which forces the geometric parameter m to satisfy $m \leq d$; and a direct geodesic Poincaré inequality that improves the spectral gap used in the $\mu = 1$ branch by a factor of eight. [LLM] The remaining new step is an analytic sharpening of Kivva’s $\mu = 2$ multiplicity argument. It uses an exact recurrence for relative drops of the standard sequence and a direct concentration bound for the dominant distance sphere. [LLM] Unlike earlier exploratory versions, the proof uses no finite tuple enumeration and no post-2021 classification theorem. [LLM]

Verification status. The proof below incorporates a line-by-line source-interface and algebra audit. Every external input is identified by theorem number using the journal numbering. Section 9 records the relevant arXiv–journal numbering concordance, and the accompanying file `verify_scalar_inequalities_strengthened.py` checks the new scalar inequalities appearing below and the displayed polynomial certificates with exact rational arithmetic. [LLM/editorial] This is nevertheless not independent peer review: the manuscript should not be cited as an established result until a specialist has checked the new $\mu = 2$ argument and the source interfaces. [LLM/editorial]

Provenance convention. Every mathematical assertion and every displayed equation is followed by a provenance label. [PS, ...] refers to Pyber–Skresanov [1]; [K, ...] refers to Kivva [2]; both use the journal numbering unless explicitly stated otherwise. [LLM] means that the assertion is proved, defined, or algebraically derived in this manuscript rather than copied from an identified source; it is not by itself a claim of conceptual novelty. A combined label, such as [PS, Proposition 2.8; LLM weighted modification], records both the imported input and the new modification. [LLM/editorial]

Scope of this strengthened audit version. This version makes one numerical change to the source-annotated proof: the coefficient 12 is replaced by the exact rational constant $C_0 = 5673/500 = 11.346$. The change is confined to the parameter definitions, the scalar inequalities derived from them, and the exact Johnson-threshold comparison below; no structural lemma, classification theorem, or source interface is changed. [LLM/editorial; strengthened scalar retuning] The more elaborate diameter-dependent proposals approaching a denominator $4d^3$ are deliberately not included, because their piecewise optimization would make the source audit substantially less readable. [LLM/editorial]

1. STATEMENT AND PARAMETERS

Let X be a primitive distance-regular graph on n vertices, with valency k , diameter $d \geq 3$, intersection numbers b_i, c_i , and

$$\lambda = a_1, \quad \mu = c_2. \quad [\text{LLM}]$$

Write k_i for the size of a distance- i sphere, θ for the second largest adjacency eigenvalue, and $-m$ for the smallest eigenvalue, so $m > 0$. [LLM/notation]

Theorem 1.1. *Either X is a Johnson graph or a Hamming graph, or*

$$(1) \quad \text{motion}(X) \geq \frac{500n}{5673d^3} = \frac{n}{11.346 d^3}. \quad [\text{LLM; strengthened scalar retuning}]$$

Distance-regular graphs of valency 2 are cycles. [PS, Section 2.1] If $k = 2$, every nonidentity automorphism of a cycle moves at least $n - 2$ vertices; since $d \geq 3$, this is greater than $500n/(5673d^3)$. [LLM; strengthened scalar retuning] Henceforth we assume

$$k > 2.$$

[PS, standing assumption after Section 2.1; LLM cycle dispatch]

Set [LLM/definition; strengthened scalar retuning]

$$(2) \quad C_0 = \frac{5673}{500} = 11.346, \quad \gamma = \frac{1}{C_0 d^3} = \frac{500}{5673 d^3}, \quad \varepsilon = \frac{2}{C_0 d^3 - 1} = \frac{1000}{5673 d^3 - 500}.$$

[LLM; strengthened scalar retuning]

The parameters satisfy the exact closure identity

$$(3) \quad \frac{\varepsilon(1 - \gamma)}{2} = \gamma.$$

[LLM; strengthened scalar retuning]

We shall repeatedly use

$$(4) \quad \gamma < \frac{1}{2}, \quad \varepsilon < \frac{2}{11d^3}, \quad d\varepsilon < \frac{1}{50}.$$

[LLM; strengthened scalar retuning]

The last inequality is strongest at $d = 3$, where

$$d\varepsilon = \frac{3000}{152671} < \frac{1}{50}.$$

[LLM; strengthened scalar retuning]

2. ADJACENT DISTINGUISHERS AND AUTOMORPHISM SUPPORT

For vertices u, v , put

$$D(u, v) = \{z : \text{dist}(u, z) \neq \text{dist}(v, z)\}.$$

[LLM]

For adjacent u, v , the cardinality is constant by distance-regularity; denote it by $D(1)$. [LLM/notation]

Lemma 2.1 (Exact adjacent-pair identity). *For every distance-regular graph of diameter at least three,* [LLM]

$$(5) \quad D(1) = 2 + \frac{2}{k} \sum_{i=2}^d k_i c_i.$$

[LLM]

Consequently, [LLM]

$$(6) \quad D(1) > \frac{\mu}{k} n.$$

[LLM]

Proof. For adjacent u, v , the number of vertices at distance i from both is $p_{i,i}^1$. [LLM/notation] A double count gives the balance identity

$$k p_{i,i}^1 = k_i p_{1,i}^i = k_i a_i,$$

[LLM, double count]

so

$$D(1) = n - \frac{1}{k} \sum_{i=1}^d k_i a_i.$$

[LLM]

Using $a_i = k - b_i - c_i$ and the standard sphere recurrence $k_i b_i = k_{i+1} c_{i+1}$, together with $b_d = 0$ and $k_1 c_1 = k$, we obtain [PS, Section 2.1; K, Section 2.2, (3 γ)]

$$\sum_{i=1}^d k_i a_i = k(n - 2) - 2 \sum_{i=2}^d k_i c_i,$$

[LLM, telescoping]

which proves (5). [LLM]

We also use the comparison $c_i \leq b_j$ whenever $i + j \leq d$. Place vertices u, w, v in this order on a geodesic with $\text{dist}(u, w) = i$ and $\text{dist}(w, v) = j$. Every neighbor z of w with $\text{dist}(u, z) = i - 1$ must satisfy $\text{dist}(v, z) = j + 1$, so the c_i such neighbors form a subset of the b_j such neighbors. [LLM] In particular $c_2 \leq b_1$, hence

$$k_2 = \frac{k b_1}{c_2} \geq k_1 = k.$$

[K, Section 2.2, (3 γ); LLM]

It follows that $\sum_{i=2}^d k_i \geq (n-1)/2$. [LLM] Since $c_i \geq c_2 = \mu$ for $i \geq 2$, [PS, Section 2.1]

$$D(1) \geq 2 + \frac{\mu}{k}(n-1) > \frac{\mu}{k}n,$$

[LLM]

where the final inequality uses $\mu \leq k$. [PS, Section 2.1; LLM] $\square$

Lemma 2.2 (Support-sensitive adjacent pair). *Let $g \in \text{Aut}(X)$ be nonidentity, put $S = \text{supp}(g)$ and $\rho = |S|/n \leq 1/2$. Some $x \in S$ has at least* [PS, proof of Proposition 2.8; LLM support-sensitive form]

$$(7) \quad \frac{k}{d}(1-\rho)$$

[PS, proof of Proposition 2.8, final inequality; LLM notation]

fixed neighbors. If $\mu < k(1-\rho)/d$, then $x \sim x^g$ and [LLM]

$$(8) \quad \lambda \geq \frac{k}{d}(1-\rho), \quad D(1) \leq |S|.$$

[LLM]

Proof. We spell out the orientation convention in the support-sensitive form of Pyber–Skresanov’s Proposition 2.8. [PS, Proposition 2.8; LLM orientation bookkeeping] For a directed edge e of X , let

$$P_e = \sum_{u,v} \frac{1}{p(u,v)} \#\{L : L \text{ is a } u\text{--}v \text{ geodesic using } e\},$$

[PS, proof of Proposition 2.8]

where $p(u,v)$ is the number of u – v geodesics. [PS, proof of Proposition 2.8] Their coherent-configuration argument shows that P_e is independent of the directed edge; write the common value as P . [PS, Lemma 2.7 and proof of Proposition 2.8] Summing over the nk directed edges gives

$$nkP = \sum_{u,v} \text{dist}(u,v) \leq dn^2, \quad \text{so} \quad P \leq \frac{dn}{k}.$$

[PS, proof of Proposition 2.8]

Let

$$\delta_X^+(S) = \{(u,v) : u \in S, v \notin S, u \sim v\}.$$

[LLM/notation]

For each ordered pair $(x,y) \in S \times (V(X) \setminus S)$, every x – y geodesic uses at least one edge of $\delta_X^+(S)$. [PS, proof of Proposition 2.8] Consequently

$$|S|(n-|S|) \leq |\delta_X^+(S)|P,$$

[PS, proof of Proposition 2.8]

and hence [LLM]

$$(9) \quad |\delta_X^+(S)| \geq |S| \frac{k}{d} \frac{n-|S|}{n}.$$

[PS, proof of Proposition 2.8, final inequality]

Each undirected cut edge contributes exactly one directed edge to $\delta_X^+(S)$, so this is also the corresponding undirected cut size. Averaging the outbound cut degrees over the vertices of S proves (7). [LLM, directed/undirected conversion] Since S is the support of g , every neighbor outside S is fixed. [LLM]

Every fixed neighbor of x is also adjacent to x^g . If x and x^g were nonadjacent, the existence of a common neighbor would put them at distance two, where they have exactly μ common neighbors. This contradicts the strict hypothesis. Thus $x \sim x^g$, and the fixed common neighbors give the lower bound on λ . [LLM]

For $z \notin S$,

$$\text{dist}(x,z) = \text{dist}(x^g,z^g) = \text{dist}(x^g,z),$$

[LLM]

so no fixed vertex distinguishes x and x^g . Hence $D(1) \leq |S|$. [LLM] $\square$

3. A DIRECT GEODESIC POINCARÉ INEQUALITY

The next proposition is the self-contained spectral improvement used in the $\mu = 1$ branch. [LLM]

Proposition 3.1 (Geodesic spectral-gap bound). *Let G be the graph of a connected symmetric basis relation of valency k in a homogeneous coherent configuration on n points. If D is its diameter and θ_1 its second largest adjacency eigenvalue, then* [LLM]

$$(10) \quad k - \theta_1 \geq \frac{n^2 k}{\sum_{x,y \in V(G)} \text{dist}(x,y)^2} \geq \frac{k}{D^2}.$$

[LLM]

For a distance-regular graph, [LLM]

$$(11) \quad k - \theta_1 \geq \frac{nk}{\sum_{i=0}^D k_i i^2}.$$

[LLM]

Proof. For ordered vertices x, y , let $p(x, y)$ be the number of geodesics from x to y . For a directed edge e , define [PS, proof of Proposition 2.8; LLM weighted modification]

$$Q_e = \sum_{x,y} \frac{\text{dist}(x,y)}{p(x,y)} \# \{P : P \text{ is an } x\text{-}y \text{ geodesic containing } e\}.$$

[PS, proof of Proposition 2.8; LLM distance weight]

The intersection-number argument in Pyber–Skresanov’s proof of Proposition 2.8 shows that Q_e is independent of e . After fixing the three basis relations containing (x, z) , (w, y) , and (x, y) for $e = (z, w)$, the number of relevant pairs and geodesics is independent of (z, w) by their Lemma 2.7; the extra weight $\text{dist}(x, y)/p(x, y)$ depends only on the basis relation containing (x, y) . Write the common value as Q . [PS, Lemma 2.7 and proof of Proposition 2.8; LLM weighted modification]

Summing over the nk directed edges gives

$$(12) \quad nkQ = \sum_{x,y} \text{dist}(x,y)^2,$$

[LLM, weighted load double count]

because every geodesic of length ℓ contributes ℓ at each of its ℓ directed edges. [LLM]

Let f have mean zero. Averaging Cauchy–Schwarz along all geodesics from x to y gives [LLM]

$$(f(x) - f(y))^2 \leq \frac{\text{dist}(x,y)}{p(x,y)} \sum_{P:x \rightarrow y} \sum_{e \in P} (\nabla_e f)^2.$$

[LLM]

After summing over ordered pairs and using the constancy of Q_e , [PS, Lemma 2.7; LLM]

$$2n\|f\|_2^2 \leq Q \sum_{e \text{ directed}} (\nabla_e f)^2 = 2Q f^\top (kI - A) f.$$

[LLM]

Taking f in the θ_1 -eigenspace gives $k - \theta_1 \geq n/Q$. Equation (12) proves the first inequality in (10); the bound $\text{dist}(x, y) \leq D$ proves the second. [LLM] In a distance-regular graph,

$$\sum_{x,y} \text{dist}(x,y)^2 = n \sum_{i=0}^D k_i i^2,$$

[LLM]

which proves (11). [LLM] $\square$

Remark 3.2. Pyber and Skresanov obtain $k - \theta_1 \geq k/(8D^2)$ by passing from their geodesic expansion estimate through Cheeger’s inequality. [PS, Proposition 2.9] Proposition 3.1 applies Cauchy–Schwarz directly to the same uniform geodesic loads and retains the squared path lengths. [LLM] No novelty claim is made for this canonical-path inequality without a broader literature search. [LLM/editorial]

4. SMALL SUPPORT FORCES DELSARTE GEOMETRY

The proof of Pyber–Skresanov’s Proposition 2.6 retains the following consequence of Metsch’s theorem. If $\lambda^2 \geq 4k\mu$, then the graph contains a clique of size at least [PS, proof of Proposition 2.6, equations (1)–(2); Metsch]

$$(13) \quad \lambda + 2 - \left(\left\lceil \frac{3k}{2(\lambda + 1)} \right\rceil - 1 \right) (\mu - 1). \quad [\text{PS, proof of Proposition 2.6, equation (2)}]$$

Proposition 4.1 (Structural reduction). *Suppose that a nonidentity automorphism has support density $\rho < \gamma$. Then* [LLM]

$$(14) \quad \mu < \gamma k, \quad \lambda > \frac{1 - \gamma}{d} k, \quad k > C_0 d^3, \quad [\text{LLM}]$$

and X is Delsarte-geometric with smallest eigenvalue $-m$ satisfying [LLM]

$$(15) \quad m \leq d. \quad [\text{LLM}]$$

Proof. Suppose first that $\mu \geq k/(2d)$. Let $k_i = k_{\max}$ be a largest nontrivial relation valency. Since $k_2 \geq k_1$ by Lemma 2.1, we may take $i \geq 2$. [LLM] Then

$$\frac{k_i}{k_{i-1}} = \frac{b_{i-1}}{c_i} \leq \frac{k}{\mu} \leq 2d, \quad [\text{PS, Section 2.1 sphere recurrence; LLM}]$$

so $n - k_{\max} \geq k_{i-1} \geq k_{\max}/(2d)$. [LLM] Every nontrivial relation of the primitive distance scheme has diameter at most d . [PS, Section 2.2, paragraph before Lemma 2.7] Hence Pyber–Skresanov Propositions 2.10 and 2.12 give

$$\text{motion}(X) \geq \frac{n - k_{\max}}{d}. \quad [\text{PS, Propositions 2.10 and 2.12}]$$

If $k_{\max} \geq n/2$, this is at least $n/(4d^2)$; otherwise it is greater than $n/(2d)$. Both bounds exceed γn , contradicting $\rho < \gamma$. [LLM] Therefore

$$(16) \quad \mu < \frac{k}{2d}. \quad [\text{LLM}]$$

Since $\rho < \gamma < 1/2$, Lemma 2.2 applies and gives [LLM]

$$\lambda > \frac{1 - \gamma}{d} k. \quad [\text{LLM}]$$

Together with Lemma 2.1 and $D(1) \leq |S|$, it gives [LLM]

$$\mu < \rho k < \gamma k. \quad [\text{LLM}]$$

As $\mu \geq 1$, this also implies $k > 1/\gamma = C_0 d^3$. [LLM; strengthened scalar retuning]

Put $\alpha = (1 - \gamma)/d$. We need three elementary inequalities: [LLM]

$$(17) \quad \alpha^2 > 4\gamma, \quad \alpha - \frac{3\gamma}{2\alpha} > \frac{1}{d+1}, \quad (d+1)^2 \gamma < \alpha. \quad [\text{LLM}]$$

For the first, $1 - \gamma > 1/2$ and $C_0 d > 16$ give [LLM; strengthened scalar retuning]

$$\alpha^2 > \frac{1}{4d^2} > \frac{4}{C_0 d^3} = 4\gamma. \quad [\text{LLM; strengthened scalar retuning}]$$

For the second, [LLM; strengthened scalar retuning]

$$\begin{aligned} d^2 \left(\alpha - \frac{1}{d+1} - \frac{3\gamma}{2\alpha} \right) &= \frac{d}{d+1} - \frac{1}{C_0 d^2} - \frac{3}{2C_0(1-\gamma)} \\ &> \frac{3}{4} - \frac{1}{99} - \frac{3}{11} > 0. \end{aligned} \quad [\text{LLM; strengthened scalar retuning}]$$

For the third, [LLM; strengthened scalar retuning]

$$d(d+1)^2 \gamma = \frac{(d+1)^2}{C_0 d^2} \leq \frac{16}{9C_0} < \frac{16}{99} < \frac{1}{2} < 1 - \gamma. \quad [\text{LLM; strengthened scalar retuning}]$$

The first inequality and (14) give $\lambda^2 > 4k\mu$, so the sourced Metsch bound (13) applies. Let L be the resulting clique size. Since $\lceil x \rceil - 1 < x$ for every real x , [PS, proof of Proposition 2.6, equation (2); LLM parameter substitution]

$$\begin{aligned} L - 1 &\geq \lambda + 1 - \left(\left\lceil \frac{3k}{2(\lambda + 1)} \right\rceil - 1 \right) (\mu - 1) \\ &> \alpha k - \frac{3k}{2(\lambda + 1)} \mu > k \left(\alpha - \frac{3\gamma}{2\alpha} \right) > \frac{k}{d + 1}. \end{aligned} \quad [\text{LLM}]$$

The Delsarte clique bound gives $L \leq 1 + k/m$, and therefore $m < d + 1$. [PS, Lemma 2.2; Delsarte] Finally,

$$m^2 \mu < (d + 1)^2 \gamma k < \alpha k < \lambda. \quad [\text{LLM}]$$

By the Bang–Koolen criterion, X is Delsarte-geometric. [PS, Proposition 2.5; Koolen–Bang] In a Delsarte geometry m is the integer number of Delsarte cliques through each vertex, so $m \leq d$. [PS, Lemma 2.3; LLM rounding] $\square$

5. THE TRANSITION AND SPECTRAL BRANCHES

Assume from now on that a nonidentity automorphism has support density $\rho < \gamma$. Proposition 4.1 is in force. [LLM]

Lemma 5.1 (Transition without a diameter loss). *If*

$$b_j \geq \varepsilon k, \quad c_{j+1} \geq \varepsilon k \quad [\text{LLM}]$$

for some $1 \leq j \leq d - 1$, then $|S| > \varepsilon n > \gamma n$. [LLM]

Proof. For $i \leq j$, monotonicity gives $b_i \geq \varepsilon k$; for $i \geq j + 1$, it gives $c_i \geq \varepsilon k$. [PS, Section 2.1] Thus $a_i \leq (1 - \varepsilon)k$ for every $i \geq 1$, and [LLM]

$$D(1) = n - \frac{1}{k} \sum_{i=1}^d k_i a_i \geq n - (1 - \varepsilon)(n - 1) > \varepsilon n. \quad [\text{LLM}]$$

Now use $D(1) \leq |S|$ from Lemma 2.2. [LLM] $\square$

If no transition occurs, let t be the least index with $b_t \leq \varepsilon k$, using the convention $b_d = 0$. [LLM] The inequality $2\lambda \leq k + \mu$ gives [PS, Lemma 2.4]

$$b_1 = k - \lambda - 1 \geq \frac{k - \mu - 2}{2} > \frac{1 - 3\gamma}{2} k > \frac{k}{3}, \quad [\text{LLM}]$$

where $1/k < \gamma$ was used. Hence $t \geq 2$. Since $b_{t-1} > \varepsilon k$ and no transition occurs, [LLM]

$$(18) \quad b_t \leq \varepsilon k, \quad c_t < \varepsilon k. \quad [\text{LLM}]$$

Lemma 5.2 (High second eigenvalue). *Under these assumptions,* [LLM]

$$(19) \quad \theta \geq (1 - \varepsilon)b_1. \quad [\text{LLM}]$$

Proof. Suppose instead that $\theta < (1 - \varepsilon)b_1$. Since $m \leq d$, $b_1 > k/3$, $k > C_0 d^3$, and $\varepsilon < 1/2$, [LLM; strengthened scalar retuning]

$$(1 - \varepsilon)b_1 > \frac{k}{6} > \frac{C_0 d^3}{6} > d \geq m. \quad [\text{LLM; strengthened scalar retuning}]$$

Thus the zero-weight spectral radius $\xi = \max\{\theta, m\}$ satisfies $\xi < (1 - \varepsilon)b_1$. [PS, paragraph before Proposition 2.13; LLM]

Moreover $\lambda > \mu$, because $(1 - \gamma)/d > \gamma$. Two distinct vertices have λ , μ , or 0 common neighbors according as their distance is 1, 2, or at least 3; hence the common-neighbor maximum over distinct pairs is λ . [LLM] Babai's spectral motion bound therefore gives [PS, Proposition 2.13]

$$\begin{aligned} \rho &\geq \frac{k - \xi - \lambda}{k} > \frac{1 + \varepsilon b_1}{k} \\ &= \frac{1 - \varepsilon}{k} + \varepsilon \frac{k - \lambda}{k} \geq \frac{1 - \varepsilon}{k} + \frac{\varepsilon}{2} \left(1 - \frac{\mu}{k} \right) > \frac{\varepsilon}{2} (1 - \gamma) = \gamma, \end{aligned} \quad [\text{LLM}]$$

contradicting $\rho < \gamma$. The last equality is (3). [LLM] $\square$

6. THE PUBLISHED JOHNSON AND $\mu = 1$ ENDGAMES

Proposition 6.1 (The case $\mu \geq 3$). *Under (18) and (19), if $\mu \geq 3$, then X is a Johnson graph.* [PS, Proposition 2.20; K, Theorem 1.2 and Proposition 3.6; LLM assembly]

Proof. If the neighborhood graphs are disconnected, Pyber–Skresanov Proposition 2.20 gives [PS, Proposition 2.20]

$$\theta + 1 \leq \frac{5}{7}b_1,$$

[PS, Proposition 2.20]

contradicting (19), since $\varepsilon < 2/7$. Hence the neighborhood graphs are connected. [LLM]

Kivva defines the Johnson threshold in Proposition 3.6 as follows. Let [K, Theorem 3.5 and Proposition 3.6]

$$p(x) = x^2(x^2 - 1)^2(x^2 - 3)(x^2 - 4) - 1,$$

[K, Theorem 3.5; LLM notation]

let $\vartheta_1 < -2$ be its smallest root, and put [K, Theorem 3.5 and Proposition 3.6]

$$\varepsilon_K = \frac{-2 - \vartheta_1}{-1 - \vartheta_1}.$$

[K, Proposition 3.6]

The threshold has the exact rational lower bound [LLM; strengthened scalar retuning]

$$\varepsilon_K > \frac{3}{458}.$$

[K, Theorem 3.5 and Proposition 3.6; LLM exact root bracket]

Indeed, writing $y = (2 + s)^2 \geq 4$, the function $y(y - 1)^2(y - 3)(y - 4) - 1$ is strictly increasing in y , so $p(-2 - s)$ is strictly increasing for $s \geq 0$. [LLM] Exact evaluation after clearing the positive denominator gives [LLM]

$$455^{10}p\left(-\frac{913}{455}\right) = -12841664057813389062001 < 0.$$

[LLM exact rational certificate]

Thus $-2 - \vartheta_1 > 3/455$, which is equivalent to the displayed threshold. [LLM] Moreover, since $d \geq 3$, [LLM; strengthened scalar retuning]

$$\varepsilon \leq \frac{1000}{152671} < \frac{3}{458} < \varepsilon_K, \quad \frac{3}{458} - \frac{1000}{152671} = \frac{13}{69923318}.$$

[LLM exact rational certificate]

Kivva's Theorem 1.2 states that a Delsarte-geometric distance-regular graph with $\mu \geq 2$, a connected neighborhood graph, [K, Theorem 1.2]

$$\theta + 1 > (1 - \varepsilon_K)b_1, \quad k \geq \max\{m^3, 29\},$$

[K, Theorem 1.2]

is Johnson. [K, Theorem 1.2] Equation (19) and $\varepsilon < \varepsilon_K$ give [LLM; strengthened scalar retuning]

$$\theta + 1 \geq (1 - \varepsilon)b_1 + 1 > (1 - \varepsilon_K)b_1.$$

[LLM; strengthened scalar retuning]

Also [LLM; strengthened scalar retuning]

$$k > C_0 d^3 \geq C_0 m^3 > m^3, \quad k > C_0 3^3 > 29.$$

[LLM; strengthened scalar retuning]

Thus all hypotheses of Kivva's theorem hold. [LLM, hypothesis check] $\square$

Proposition 6.2 (The case $\mu = 1$). *Under Proposition 4.1, the case $\mu = 1$ contradicts $\rho < \gamma$.* [PS, Propositions 2.14–2.15; LLM assembly]

Proof. Proposition 3.1 gives [LLM]

$$\theta \leq k \left(1 - \frac{1}{d^2}\right).$$

[LLM]

Put $\eta = 1/d^2$. Since $d \geq 3$, $0 < \eta < 1/2$. Moreover [LLM]

$$m \leq d < k \left(1 - \frac{1}{d^2}\right),$$

[LLM]

because $k > C_0 d^3$. [LLM; strengthened scalar retuning] Hence the zero-weight spectral radius satisfies [PS, paragraph before Proposition 2.13; LLM]

$$\xi = \max\{\theta, m\} \leq k(1 - \eta).$$

[LLM]

If $m \geq 3$, then [LLM]

$$k > C_0 d^3 > 4md^2 = \frac{4m}{\eta}, \quad k > m^2. \quad [\text{LLM}]$$

Pyber–Skresanov Proposition 2.14 therefore yields [PS, Proposition 2.14; LLM hypothesis check]

$$\text{motion}(X) \geq \frac{\eta n}{4} = \frac{n}{4d^2} > \gamma n. \quad [\text{PS, Proposition 2.14; LLM substitution}]$$

If $m < 3$, geometric integrality gives $m = 1$ or 2 . [PS, Lemma 2.3] The case $m = 1$ is impossible: every vertex would lie in exactly one Delsarte clique, and connectedness would force the graph to be complete, contrary to $d \geq 3$. Thus $m = 2$. [LLM] Here $k > C_0 d^3 > 4$, so the remaining valency hypothesis of Pyber–Skresanov Proposition 2.15 is satisfied, and that proposition gives $\text{motion}(X) \geq n/16 > \gamma n$. [PS, Proposition 2.15; LLM hypothesis check] $\square$

7. AN ANALYTIC HAMMING-STABILITY LEMMA

We now handle the only branch not covered by a published endgame with the desired parameter scale. [LLM]

Proposition 7.1 (Sharpened $\mu = 2$ endgame). *Let X be a Delsarte-geometric distance-regular graph of diameter $d \geq 3$, with smallest eigenvalue $-m$, where $m \leq d$, and $\mu = 2$. Let* [LLM]

$$0 < \varepsilon = \frac{2}{C_0 d^3 - 1}. \quad [\text{LLM}]$$

Suppose that, for some $2 \leq t \leq d$, [LLM]

$$(20) \quad b_t \leq \varepsilon k, \quad c_t < \varepsilon k, \quad \theta \geq (1 - \varepsilon)b_1, \quad k > C_0 d^3. \quad [\text{LLM}]$$

Then X is a Hamming graph. [LLM]

Proof. The geometric identities of Kivva’s Lemma 2.17 are [K, Lemma 2.17 (journal)]

$$(21) \quad c_i = \tau_i \psi_{i-1}, \quad b_i = (m - \tau_i) \left(\frac{k}{m} + 1 - \psi_i \right). \quad [\text{K, Lemma 2.17 (journal)}]$$

Here $\psi_0 = 1$, so $c_1 = 1 = \tau_1 \psi_0$ gives [K, equations (1)–(2) and Lemma 2.17 (journal); LLM]

$$\tau_1 = 1. \quad [\text{LLM}]$$

Kivva’s Lemma 2.18 gives $\tau_2 \geq \psi_1$. Since $\mu = \tau_2 \psi_1 = 2$, we have [K, Lemmas 2.17–2.18 (journal); LLM integrality]

$$(22) \quad \tau_2 = 2, \quad \psi_1 = 1, \quad b_1 = \frac{m-1}{m}k, \quad \lambda = \frac{k}{m} - 1. \quad [\text{K, Lemmas 2.17–2.18 (journal); LLM}]$$

In particular, every neighborhood graph is a disjoint union of m cliques. [K, Lemma 2.20 (journal)] Since X is geometric and $\mu = 2$, an induced quadrangle exists, so Terwilliger’s inequality is available wherever it is used below. [K, Lemma 3.10 and Theorem 2.6]

Since

$$\varepsilon = \frac{2}{C_0 d^3 - 1} < \frac{1}{d^2} \leq \frac{1}{m^2}, \quad [\text{LLM}]$$

Kivva’s Lemma 4.2 applies and gives [K, Lemma 4.2; LLM hypothesis check]

$$(23) \quad \tau_i < \tau_{i+1} \quad (1 \leq i \leq t-2). \quad [\text{K, Lemma 4.2}]$$

The parameters are integral, so $\tau_1 = 1$ and (23) imply $\tau_i \geq i$ for $i \leq t-1$. [K, equation (2) defining τ_i ; LLM] Put

$$(24) \quad r = m - \tau_{t-1}. \quad [\text{LLM}]$$

Since $b_{t-1} > 0$, (21) gives $\tau_{t-1} \leq m-1$; together with $\tau_{t-1} \geq t-1$, [K, Lemma 2.17 (journal); LLM]

$$(25) \quad 1 \leq r \leq m - t + 1, \quad 2 \leq t \leq m. \quad [\text{LLM}]$$

Let $(u_i)_{i=0}^d$ be the standard sequence for θ . [K, Definition 2.9] Define the relative drops

$$(26) \quad y_i = \frac{u_{i-1} - u_i}{u_{i-1}} \quad [\text{LLM}]$$

whenever $u_{i-1} > 0$. [LLM/definition] The standard-sequence recurrence gives the exact identity [K, Section 2.2 recurrence; LLM algebra]

$$(27) \quad y_{i+1} = \frac{k - \theta + c_i y_i / (1 - y_i)}{b_i} \quad [\text{K, Section 2.2 standard-sequence recurrence; LLM change of variables}]$$

Set [LLM/definition]

$$(28) \quad B = 1 + (m-1)\varepsilon, \quad A = \frac{1 + (5m/3 - 1)\varepsilon}{1 - m\varepsilon}, \quad \delta = A - 1 = \frac{(8m-3)\varepsilon}{3(1-m\varepsilon)}. \quad [\text{LLM}]$$

From (22) and the eigenvalue hypothesis, [K, Definition 2.9 gives $u_1 = \theta/k$; LLM]

$$(29) \quad k - \theta \leq \frac{k}{m}B, \quad u_1 = \frac{\theta}{k} \geq (1-\varepsilon)\frac{m-1}{m} =: u_*. \quad [\text{K, Definition 2.9; LLM}]$$

The small-parameter estimates (4) and $m \leq d$ imply [LLM]

$$(30) \quad m\varepsilon < \frac{1}{50}, \quad A < \frac{6}{5}, \quad \delta < 3d\varepsilon. \quad [\text{LLM}]$$

Indeed, writing $x = m\varepsilon < 1/50$, [LLM]

$$A \leq \frac{1 + 5x/3}{1-x} < \frac{155}{147} < \frac{6}{5}, \quad \delta < \frac{8d\varepsilon}{3(1-x)} < \frac{400}{147}d\varepsilon < 3d\varepsilon. \quad [\text{LLM}]$$

At $i = 1$, using $c_1 = 1$, $b_1 = (m-1)k/m$, $1/k < \gamma$, and (29), [K, Lemma 2.17 (journal) and Section 2.2 recurrence; LLM]

$$y_2 \leq \frac{B}{m-1} + \frac{m\gamma}{m-1} \frac{1-u_*}{u_*} \quad [\text{LLM}]$$

and hence [LLM]

$$(31) \quad y_2 \leq \frac{1}{m-1} \left(B + \frac{m\gamma B}{(m-1)(1-\varepsilon)} \right). \quad [\text{LLM}]$$

Since [LLM]

$$\frac{B}{m-1} = \frac{1}{m-1} + \varepsilon \leq 1 + \varepsilon, \quad [\text{LLM}]$$

while $(1+\varepsilon)/(1-\varepsilon) < 2$ and $2\gamma < \varepsilon$, we have [LLM]

$$\frac{\gamma B}{(m-1)(1-\varepsilon)} < \varepsilon. \quad [\text{LLM}]$$

Furthermore [LLM]

$$A - B = \frac{m\varepsilon(B+2/3)}{1-m\varepsilon} > m\varepsilon. \quad [\text{LLM}]$$

Hence [LLM]

$$(32) \quad 0 \leq y_2 < \frac{A}{m-1}. \quad [\text{LLM}]$$

If $t = 2$, the product below is empty and only the already positive entry u_1 is used. If $t \geq 3$, then $m \geq 3$, so (32) and $A < 6/5$ give $y_2 < 3/5 < 1$ and hence $u_2 > 0$. [LLM, edge-case check]

We claim inductively that [LLM]

$$(33) \quad 0 \leq y_{i+1} \leq \frac{A}{m - \tau_i} \quad (1 \leq i \leq t-2). \quad [\text{LLM}]$$

The base is (32). Suppose $2 \leq i \leq t-2$ and the preceding bound holds. Strict growth gives [K, Lemma 4.2; LLM]

$$m - \tau_{i-1} \geq r + t - i \geq 3, \quad [\text{LLM}]$$

so $y_i \leq A/3 < 2/5$ and therefore $y_i/(1 - y_i) < 2/3$. [LLM] Also $c_i \leq c_t < \varepsilon k$ by monotonicity, and $\psi_i \leq c_{i+1} \leq c_t$ by (21). [PS, Section 2.1; K, Lemma 2.17 (journal)] Thus

$$b_i \geq (m - \tau_i) \frac{k}{m} (1 - m\varepsilon).$$

[K, Lemma 2.17 (journal); LLM]

Equation (27) now gives [LLM]

$$y_{i+1} \leq \frac{B + (2m/3)\varepsilon}{(m - \tau_i)(1 - m\varepsilon)} = \frac{A}{m - \tau_i}.$$

[LLM]

This proves the claim and, in particular, proves that all standard-sequence entries used below are positive. [LLM] Using (23), for $2 \leq i \leq t - 1$ we have $m - \tau_{i-1} \geq r + t - i$. [K, Lemma 4.2; LLM] Therefore

$$\begin{aligned} u_{t-1} &= u_1 \prod_{i=2}^{t-1} (1 - y_i) \\ &\geq u_1 \prod_{s=r+1}^{r+t-2} \left(1 - \frac{A}{s}\right) \\ &= u_1 \frac{r}{r+t-2} \prod_{s=r+1}^{r+t-2} \left(1 - \frac{\delta}{s-1}\right) \\ &\geq u_1 \frac{r}{r+t-2} (1 - \delta H_{t-2}). \end{aligned} \tag{34}$$

[LLM]

where $H_j = 1 + 1/2 + \dots + 1/j$ and $H_0 = 0$. [LLM/notation] The final inequality is the elementary product bound $\prod (1 - x_i) \geq 1 - \sum x_i$. [LLM]

We next show that the t -th sphere contains almost all vertices. [LLM] Put

$$q = \frac{m\varepsilon}{1 - m\varepsilon} < 1. \tag{35}$$

[LLM]

For $2 \leq i \leq t$, (21) and the sphere recurrence give [K, Lemma 2.17 (journal); K, Section 2.2, (3r)]

$$\frac{k_{i-1}}{k_i} = \frac{c_i}{b_{i-1}} \leq \frac{q}{m - \tau_{i-1}} \leq q.$$

[K, Lemma 2.17 (journal) and Section 2.2, (3r); LLM]

The remaining first ratio is $k_0/k_1 = 1/k < \gamma < q$; here $q > m\varepsilon \geq 2\varepsilon > 4\gamma$. Thus the entire left tail has mass at most $k_t q / (1 - q)$. [LLM]

For $t \leq i \leq d - 1$, monotonicity gives $b_i \leq b_t \leq \varepsilon k$. [PS, Section 2.1] Since $m - \tau_i \geq 1$, the second identity in (21) implies [K, Lemma 2.17 (journal); LLM]

$$\psi_i \geq \frac{k}{m} + 1 - \varepsilon k > k \left(\frac{1}{m} - \varepsilon \right).$$

[K, Lemma 2.17 (journal); LLM]

Consequently [K, Section 2.2, (3r); LLM]

$$\frac{k_{i+1}}{k_i} = \frac{b_i}{c_{i+1}} \leq \frac{\varepsilon k}{\psi_i} \leq q,$$

[K, Section 2.2, (3r); K, Lemma 2.17 (journal); LLM]

so the right tail also has mass at most $k_t q / (1 - q)$. [LLM] Hence

$$\frac{n}{k_t} \leq 1 + \frac{2q}{1 - q} = \frac{1 + q}{1 - q}, \quad k_t \geq (1 - 2m\varepsilon)n. \tag{36}$$

[LLM]

Since $\psi_{t-1} \geq 1$, the second identity in (21) gives $b_{t-1} \leq rk/m$. [K, Lemma 2.17 (journal); LLM] Combining (29), (34), and (36), we obtain [LLM]

$$k_{t-1} u_{t-1}^2 \geq \frac{n}{k} FR, \tag{37}$$

[LLM]

where [LLM/definition]

$$F = (1 - 2m\varepsilon)(1 - \varepsilon)^2(1 - \delta H_{t-2})^2, \quad R = \frac{c_t r (m - 1)^2}{m(r + t - 2)^2}. \tag{38}$$

[LLM]

We now prove that, outside one endpoint, R provides a uniform gain: [LLM]

$$(39) \quad R \geq \frac{m+1}{m}. \quad [\text{LLM}]$$

By monotonicity and (21), [PS, Section 2.1; K, Lemma 2.17 (journal)]

$$c_t \geq c_{t-1} = \tau_{t-1} \psi_{t-2} \geq \tau_{t-1} \geq t-1. \quad [\text{PS, Section 2.1; K, Lemmas 2.17 and 4.2 (journal numbering); LLM}]$$

If $c_t = t-1$, equality throughout forces [LLM]

$$\tau_{t-1} = t-1, \quad c_t = c_{t-1}. \quad [\text{LLM}]$$

Kivva's Corollary 2.8 and Terwilliger's inequality then give $4 \leq t \leq m-1$. [K, Corollary 2.8 and Theorem 2.6; LLM equality analysis] Thus $r = m - t + 1$ and

$$R = \frac{(t-1)(m-t+1)}{m} \geq \frac{m+1}{m}; \quad [\text{LLM}]$$

indeed, the two factors sum to m , the first is at least 3, the second at least 2, and their product is at least $2(m-2) \geq m+1$. [LLM]

Suppose next that $c_t \geq t$ and $t < m$. For fixed m, t , the function [LLM]

$$f(r) = \frac{r}{(r+t-2)^2} \quad [\text{LLM}]$$

has no interior minimum on $1 \leq r \leq m-t+1$, so it is enough to check the endpoints. [LLM] At $r = 1$, [LLM]

$$R \geq \frac{t(m-1)^2}{m(t-1)^2} \geq \frac{m+1}{m}, \quad [\text{LLM}]$$

because $t/(t-1)^2$ decreases with t and [LLM]

$$(m-1)^3 - (m+1)(m-2)^2 = 3m-5 > 0. \quad [\text{LLM}]$$

At $r = m-t+1$, [LLM]

$$R \geq \frac{t(m-t+1)}{m} \geq \frac{m+1}{m}, \quad [\text{LLM}]$$

since the concave function $t(m+1-t)$ has endpoint minimum $2(m-1) \geq m+1$ on $2 \leq t \leq m-1$. [LLM]

Finally suppose $t = m$, so $r = 1$ and $R = c_t/m$. [LLM] If $t < d$, the equality $c_t = t = m$ is impossible. Strict growth gives $c_{t-1} \geq t-1 = m-1$, while $b_t \geq 1$; hence $c_{t-1} - c_t + b_t \geq 0$. [K, Lemma 4.2; LLM] Terwilliger's inequality and (21) then give [K, Theorem 2.6 and Lemma 2.17 (journal); LLM]

$$\frac{k}{m} \geq b_{t-1} \geq c_{t-1} - c_t + b_t + \lambda + 2 \geq \frac{k}{m} + 1, \quad [\text{K, Theorem 2.6 and Lemma 2.17 (journal); LLM}]$$

a contradiction. Hence $c_t \geq m+1$. If $t = m = d$, then every nonendpoint case also has $c_t \geq m+1$. This proves (39) unless [LLM]

$$(40) \quad c_t = t = m = d. \quad [\text{LLM}]$$

It remains to show that the loss factor satisfies [LLM]

$$(41) \quad F > \frac{m}{m+1}. \quad [\text{LLM}]$$

Let $z = \delta H_{t-2}$. We use the elementary bound [LLM]

$$(42) \quad H_{d-2} \leq \frac{d}{2} \quad (d \geq 3). \quad [\text{LLM}]$$

It holds at $d = 3$, and the induction step follows from [LLM]

$$H_{d-1} = H_{d-2} + \frac{1}{d-1} \leq \frac{d}{2} + \frac{1}{2} = \frac{d+1}{2}. \quad [\text{LLM}]$$

Thus, by (30), [LLM]

$$0 \leq z < \frac{3}{2}d^2\varepsilon. \quad [\text{LLM}]$$

Moreover, [LLM]

$$\frac{3}{2}d^2\varepsilon = \frac{3d^2}{C_0d^3 - 1} < 1 \quad (d \geq 3), \quad [\text{LLM}]$$

so $0 \leq z < 1$. Applying $\prod_j (1 - x_j) \geq 1 - \sum_j x_j$ to the five factors defining F gives [LLM]

$$F \geq 1 - (2m\varepsilon + 2\varepsilon + 2z). \quad [\text{LLM}]$$

Since $m \leq d$, [LLM]

$$2m\varepsilon + 2\varepsilon + 2z < (3d^2 + 2d + 2)\varepsilon < \frac{1}{d+1} \leq \frac{1}{m+1}. \quad [\text{LLM}]$$

After multiplying by 500, the middle strict inequality is equivalent to [LLM; strengthened scalar retuning]

$$2673d^3 - 5000d^2 - 4000d - 2500 > 0.$$

[LLM; strengthened scalar retuning]

A completely explicit certificate is obtained by writing $d = x + 3$: [LLM]

$$2673d^3 - 5000d^2 - 4000d - 2500 = 2673x^3 + 19057x^2 + 38171x + 12671 > 0.$$

[LLM; strengthened scalar retuning]

This proves (41) without a finite case split. [LLM]

Outside the endpoint (40), equations (39) and (41) give $FR > 1$. Hence (37) yields [LLM]

$$k_{t-1}u_{t-1}^2 > \frac{n}{k}. \quad [\text{LLM}]$$

By Biggs' multiplicity formula, the multiplicity f_1 of θ satisfies $f_1 < k$. [K, Theorem 2.10; LLM one-term bound] From (20), $\theta \geq (1 - \varepsilon)b_1 > 0$, so $\theta + 1 > 0$. [LLM] Terwilliger's local-eigenvalue theorem then puts the eigenvalue [K, Theorem 4.1; LLM hypothesis check]

$$-1 - \frac{b_1}{\theta + 1} < -1 \quad [\text{K, Theorem 4.1; LLM strict inequality}]$$

in every neighborhood graph. [K, Theorem 4.1] This is impossible because a disjoint union of cliques has no eigenvalue below -1 . [K, Lemma 2.20 (journal); LLM spectral observation] Therefore the endpoint (40) must hold. [LLM]

At the endpoint, $t = m = d$. We have $\tau_1 = 1$, and (23) gives the strict integer chain [K, Lemma 4.2; LLM]

$$1 = \tau_1 < \tau_2 < \cdots < \tau_{d-1}. \quad [\text{LLM}]$$

For $i < d$, $b_i > 0$ and (21) give $\tau_i \leq m - 1 = d - 1$, while $\tau_d = m = d$. Hence [K, Lemma 2.17 (journal); LLM integer-chain argument]

$$\tau_i = i \quad (1 \leq i \leq d). \quad [\text{LLM}]$$

Now $c_d = d = \tau_d \psi_{d-1}$ gives $\psi_{d-1} = 1$. [K, Lemma 2.17 (journal); LLM] If some ψ_j were at least 2, choose j maximal with that property. Since $\psi_{d-1} = 1$ and every ψ_i is a positive integer, putting $i = j + 1$ gives $\psi_{i-1} \geq 2$ and $\psi_i = 1$ with $2 \leq i \leq d - 1$. [K, definitions of ψ_i ; LLM] Monotonicity of the c_i would give [PS, Section 2.1; K, Lemma 2.17 (journal)]

$$i + 1 = c_{i+1} = \tau_{i+1} \psi_i \geq c_i = \tau_i \psi_{i-1} \geq 2i, \quad [\text{PS, Section 2.1; K, Lemma 2.17 (journal); LLM}]$$

which is impossible. Hence $\psi_i = 1$ for every i . [LLM] Consequently

$$c_i = i, \quad b_i = (d - i) \frac{k}{d}. \quad [\text{K, Lemma 2.17 (journal); LLM endpoint substitution}]$$

In particular, $k/d = b_{d-1} \in \mathbb{Z}$, so $s = 1 + k/d$ is an integer with $s \geq 2$. [LLM] These are the intersection numbers of $H(d, s)$. [K, Hamming intersection array (journal equation (6)); LLM identification] Egawa's classification gives a Hamming or Doob graph. [K, Theorem 2.25 (journal); Egawa] A Doob graph occurs only when $s = 4$, equivalently $k = 3d$. [K, proof of Theorem 4.7] Since $k > C_0d^3 > 3d$, the Doob case is impossible, so X is Hamming. [LLM; strengthened scalar retuning] $\square$

8. COMPLETION OF THE PROOF

Proof of Theorem 1.1. Suppose that a nonidentity automorphism g has support density $\rho < \gamma$. Proposition 4.1 makes X Delsarte-geometric with $m \leq d$. Lemma 5.1 rules out a transition, so (18) holds; Lemma 5.2 gives (19). [LLM]

If $\mu \geq 3$, Proposition 6.1 makes X Johnson. If $\mu = 2$, Proposition 7.1 makes X Hamming. If $\mu = 1$, Proposition 6.2 contradicts $\rho < \gamma$. These cases exhaust the positive integer μ . Therefore every graph outside the Johnson and Hamming families has motion at least

$$\gamma n = \frac{500n}{5673d^3} = \frac{n}{11.346 d^3}.$$

[LLM; strengthened scalar retuning]

□

9. DEPENDENCY AND AUDIT LEDGER

The proof imports the following published statements. [LLM/editorial]

- (1) From Pyber–Skresanov [1]: the Delsarte clique bound and geometric integrality (Lemmas 2.2–2.3), $2\lambda \leq k + \mu$ (Lemma 2.4), Bang–Koolen geometricity (Proposition 2.5), the full Metsch expression in the proof of Proposition 2.6, the geodesic-load argument and support-sensitive final inequality in Proposition 2.8, Propositions 2.10 and 2.12–2.15, and the disconnected-neighborhood Johnson bound in Proposition 2.20. [PS, cited items]
- (2) From Kivva [2], using journal numbering: the exact Johnson threshold and connected-neighborhood characterization (Theorem 1.2, Theorem 3.5, and Proposition 3.6), Biggs’ formula (Theorem 2.10), the geometric identities and local structure (Lemmas 2.17–2.20), $c_3 > c_2$ (Corollary 2.8), Terwilliger’s inequality (Theorem 2.6), the induced-quadrangle lemma (Lemma 3.10), the strict τ_i growth lemma (Lemma 4.2), Terwilliger’s local-eigenvalue theorem (Theorem 4.1), and the Hamming/Doob endpoint classification (Theorem 2.25 and the endpoint argument in Theorem 4.7). [K, cited items]

The original sources for the principal imported tools are also listed separately: Delsarte [3], Koolen–Bang [4], Metsch [5], Biggs [6], Terwilliger [7, 8], and Egawa [9]. [LLM/editorial]

Kivva numbering concordance. In arXiv:1912.11427, journal Lemmas 2.17–2.20 appear as Lemmas 2.16–2.19, and journal Theorem 2.25 appears as Theorem 2.24. All citations in the proof use the journal numbering. [LLM/editorial, checked against both versions]

Every row of the following table is an editorial hypothesis check, not a further imported theorem. [LLM/editorial]

Interface	Hypotheses checked in the manuscript
PS standing assumption (after Section 2.1)	The valency-2 cycle case is dispatched immediately after Theorem 1.1.
PS, Proposition 2.10 interface	Every nontrivial relation of a primitive rank- $(d+1)$ distance scheme is connected and has diameter at most d .
PS, proof of Proposition 2.6	$\lambda^2 > 4k\mu$ follows from (17); the ceiling term is not discarded.
PS, Proposition 2.5	The clique bound first gives $m < d+1$, and only then is $m^2\mu < \lambda$ invoked.
PS, Proposition 2.13	The common-neighbor maximum over distinct pairs is λ because $\lambda > \mu$.
PS, Proposition 2.20; K, Theorem 1.2 and Proposition 3.6	The disconnected case has $d > 2$ and $\mu \geq 3$. In the connected case, the exact certificate gives $\varepsilon < 3/458 < \varepsilon_K$, and $k > C_0 d^3$ gives $k \geq \max\{m^3, 29\}$.
K, Lemma 4.2	$\mu = 2$, $c_t < \varepsilon k$, and $\varepsilon < 1/d^2 \leq 1/m^2$.
K, Theorem 4.1	Biggs’ formula gives the strict inequality $f_1 < k$, and (20) gives $\theta > 0$.
K, Theorem 2.25 and proof of Theorem 4.7	The strict integer chain forces $\tau_i = i$; $k/d = b_{d-1} \in \mathbb{Z}$; the only surviving case is $c_t = t = m = d$; and the Doob valency $k = 3d$ is excluded.
PS, Propositions 2.14–2.15	Delsarte geometry, Proposition 3.1, $m \leq d < k(1 - 1/d^2)$, $\xi \leq k(1 - 1/d^2)$, $k > 4md^2, m^2$, and, for $m = 2$, $k > 4$.

Remark 9.1 (Numerical margin). Kivva’s exact threshold permits the limiting uniform coefficient $C_{\text{lim}} = (1+2/\varepsilon_K)/27$. [K, Proposition 3.6; LLM scalar optimization] The rational bracket above gives the exact comparison

$$C_{\text{lim}} < \frac{919}{81} < \frac{5673}{500} = C_0, \quad \frac{5673}{500} - \frac{919}{81} = \frac{13}{40500}. \quad [\text{LLM exact rational certificate}]$$

Thus $C_0 = 11.346$ is admissible with a small explicit margin. [LLM; strengthened scalar retuning] The diameter-dependent $4d^3$ proposal is not used in this manuscript. [LLM/editorial]

Remark 9.2. The exponent 3 is intrinsic to the present structural route. Small support density ρ gives $\mu \lesssim \rho k$, while the clique argument gives $m = O(d)$ and $\lambda \gtrsim k/d$. The Bang–Koolen condition $m^2\mu < \lambda$ therefore naturally asks for $\rho = O(d^{-3})$. Improving the exponent appears to require a stronger route to geometricity, not merely better constants in the Hamming endgame. [LLM/heuristic]

REFERENCES

- [1] L. Pyber and S. V. Skresanov, *On the automorphism group of a distance-regular graph*, J. Combin. Theory Ser. B **172** (2025), 94–114; arXiv:2312.00383.
- [2] B. Kivva, *A characterization of Johnson and Hamming graphs and proof of Babai’s conjecture*, J. Combin. Theory Ser. B **151** (2021), 339–374; arXiv:1912.11427.
- [3] P. Delsarte, *An algebraic approach to the association schemes of coding theory*, Philips Res. Rep. Suppl. **10** (1973), vi+97.
- [4] J. H. Koolen and S. Bang, *On distance-regular graphs with smallest eigenvalue at least $-m$* , J. Combin. Theory Ser. B **100** (2010), 573–584.
- [5] K. Metsch, *On a characterization of bilinear forms graphs*, European J. Combin. **20** (1999), 293–306.
- [6] N. L. Biggs, *Intersection matrices for graphs*, in *Combinatorial Mathematics and its Applications* (Proc. Conf., Oxford, 1969), Academic Press, London, 1971, 15–23.
- [7] P. Terwilliger, *Distance-regular graphs with girth 3 or 4. I*, J. Combin. Theory Ser. B **39** (1985), 265–281.
- [8] P. Terwilliger, *A new feasibility condition for distance-regular graphs*, Discrete Math. **61** (1986), 311–315.
- [9] Y. Egawa, *Characterization of $H(n, q)$ by the parameters*, J. Combin. Theory Ser. A **31** (1981), 108–125.